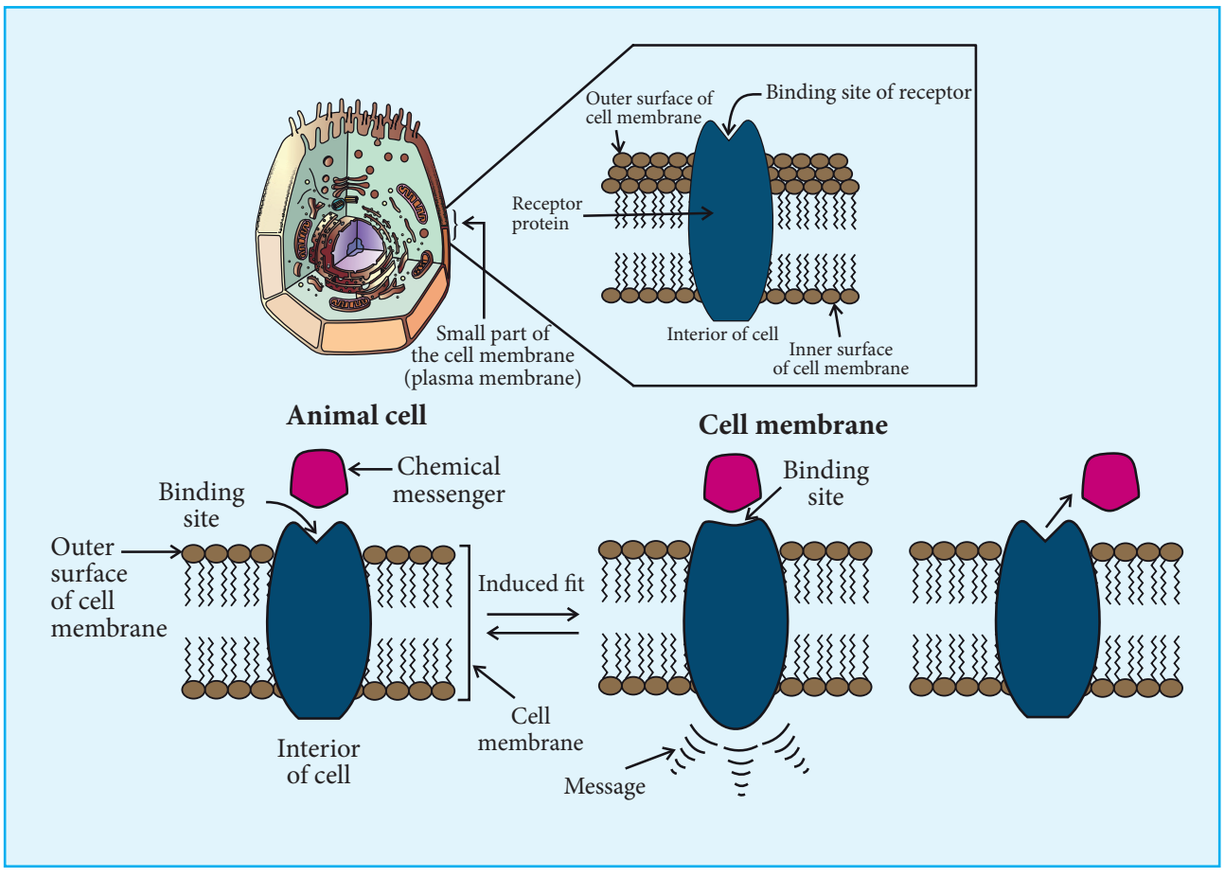
Receptor as drug targets:

Many drugs exert their physiological effects by binding to a specific molecule called a receptor whose role is to trigger a response in a cell. Most of the receptors are integrated with the cell membranes in such a way that their active site is exposed to outside region of the cell membrane. The chemical messengers, the compounds that carry messages to cells, bind to the active site of these receptors. This brings about the transfer of message into the cell. These receptors show high selectivity for one chemical messenger over the others. If we want to block a message, a drug that binds to the receptor site should inhibit its natural function. Such drugs are called **antagonists**. In contrast, there are drugs which mimic the natural messenger by switching on the receptor. These type of drugs are called **agonists** and are used when there is lack of chemical messenger.



For example, when adenosine binds to the adenosine receptors, it induces sleepiness. On the other hand, the antagonist drug caffeine binds to the adenosine receptor and makes it inactive. This results in the reduced sleepiness (wakefulness).

The agonist drug, morphine, which is used as a pain killer, binds to the opioid receptors and activates them. This supress the neuro transmitters that causes pain.

Most receptors are chiral and hence different enantiomers of a drug can have different effect

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